

Maruthavanan S V

maruthavanansvceo08@gmail.com | LinkedIn:Maruthavanan S V | Github:maruthu-05

Education

Chennai Institute Of Technology and Applied Research,
B.E in Computer Science and Engineering.

2023 – 2027

- CGPA: 8.96 / 10

Skills

Languages/Frameworks: Python, Java, C, C + +, JavaScript, Next.js, React.js, Node.js, Express.js, HTML, Tailwind CSS.

APIs and tools: Razorpay Sandbox APIs , Flask API , Supabase Auth , Storage , Database APIs , Hugging face ,Git Hub .

Databases and platform: MySQL , Firebase , MongoDB , PostgreSQL , Supabase , Pinecone , Render ,Vercel ,Railway.

Soft Skills: Problem-Solving ,Leadership ,Team work ,Critical thinking ,Communication ,Creativity ,Analytical thinking.

Projects

Database AI (DBAI)

Node.js,Exp,Python,Supabase

- A web application integrating LLM to dynamically generate and execute database queries across MySQL and MongoDB DBs.
- Designed scalable backend logic with credential encryption, and schema-aware query generation by avoiding the generation of destructive and random queries for multi-database environments and helps user to get their desired data as the outcome .
- Implemented AI-powered workflows for intelligent query synthesis, optimizing performance and ensuring compliance with security best practices like encrypted login-credentials and outputs are generated based on the user needs (query,table,csv).

Personalized AI Assistant (RAG-Based)

Springboot-Kotlin,Pinecone

- Developed a Retrieval-Augmented Generation (RAG) based personal assistant that answers context-specific questions .
- Integrated document ingestion and vector storage with Pinecone, and used Gemini API for response generation and display it.
- Enabled interactive desktop use with Python interface capable of searching,summarizing and voice interaction based on RAG.

Driver Behavior Analysis System

Opencv,Tensorflow,AI Models

- Implemented distraction of driver detection using open-source vision models (Openvino and YOLO) with storage facility.
- Features include phone usage detection,distractions , speed-limit alerts, and SOS breakdown detection for real-time recovery.
- Designed scoring logic for driver behavior evaluation , update in a database as resume and digital nudges for improvements.

NAH – Virtual Coding Bootcamp Platform

Next,Supabase,Tailwindcss

- Building a web application acts as a coding bootcamp with collaborative project tools using Next.js, Supabase, and shadcn/ui.
- Features include certificate generation, profile editing, team formation, project declarations and dynamic project display pages.
- Integrated Framer Motion for smooth transitions and hosted on Vercel along with Supabase for authentication and storage .

Competitive Programming

- **LeetCode:** Max Rating 1860 (knight), top 12.64% ,500+ problems solved in concept of DSA,Achieved highest rank 2447.
- **CodeChef:** Max Rating 1645, 2-Star, solved 250+ problems in concepts of DSA,Achieved highest rank 323 in global contest.

Certificates

- Docker foundation-Alison: Includes all the fundamental techniques and concepts of docker's basic command and execution.
- Cyberecurity Essentials -Cisco: Includes the fundamental techniques and concepts of cisco packet tracer ,linux and wireshark.
- Python for Data Science and AI-IBM-Coursera Ensures proper and deep knowledge on api,OOPS and ai-packages in python.
- Data Structures-Coursera Learned all the Data Structure (DS) concepts with a proper skill assessment questions in the course.
- Networking essentials-Cisco Covered all the basic and detailed concepts of networks and cisco packet tracer and security.

Achievements

- Shortlisted for state-level in TN-Skills competition in the field IT system administration networking conducted in 2025.
- Finalist in OneDelhi Transport Stack Open Innovation Challenge with a AI integrated behaviour analysis software solution.
- Finalist in Ideathon conducted by VIT with a green-tech web application sustainable solution on the waste management.
- Shortlisted in Nasscom AI-Agent hackathon as top 25 team across India with a full agentic model and vibe coding approach.